Garbage Collection

Community-powered clean city platform

Report garbage on a map. Organise cleanup drives. Keep our country green.

Full Project Plan

Features | Design | Tech stack | Database | Roadmap

1. Project Overview

Idea: A website where citizens select their country, state and city, log in with a phone number or email, photograph or film garbage, post it with its address, and see every report as a bin pin on a live map. People can also form city/area organisations and host cleanup events (for example Sunday or holiday drives) to spread awareness and keep the country clean.

Goals

- Make garbage visible: every report gets a pin, a photo and an address.
- Turn awareness into action: volunteer groups and cleanup events for every city.
- Give authorities clear data on hotspots so they can respond faster.
- Build a friendly, green, animated experience that people enjoy using.

Target Users

User	What they do
Citizen	Reports garbage, upvotes reports, joins events, earns points
Volunteer / Cleaner	Joins cleanup drives, uploads before/after photos
Organisation admin	Creates a city/area group, schedules events, manages members
Municipal officer	Views reports by area, updates status, marks as resolved
Platform admin	Moderates content, manages cities, sponsors and awareness content

2. User Flow

Step	Screen	What happens
1	Landing page	Animated eco hero with 'Report Garbage' and 'Join a Cleanup' buttons
2	Choose location	Select country, then state, then city (searchable dropdowns)
3	Login / Sign up	Phone number + OTP, or email + OTP/password (Google login optional)
4	City map	See all garbage reports for the chosen city as colour-coded bin pins
5	Report garbage	Take a photo/video, auto-detect GPS, confirm address, add details, post
6	Communities	Join or create an organisation for your area and host events
7	Track impact	Follow report status, points, badges and city cleanliness score

3. Core Features

A. Garbage Reporting

- Capture a photo or a short video (up to 30 seconds) directly from the camera, or upload one.
- Automatic GPS location with a draggable pin for correction.
- Address auto-filled from the pin (reverse geocoding) and editable by the user.
- Garbage type tags: plastic, household, construction debris, medical, e-waste, dead animal, drain blockage.
- Severity: small, medium or large heap. Optional anonymous posting.
- Time and location checks (EXIF data) to reduce fake or old photos.

B. Interactive Map

- Bin pin markers: **red** = reported, **orange** = in progress, **green** = cleaned.
- Marker clustering when many reports are close together, plus a heatmap of the dirtiest areas.
- Filters by type, severity, date and status; a 'Near me' button; directions to a spot.
- Tap a pin to see photos, address, votes and comments.
- Also show existing dustbins and recycling centres with separate icons.

C. Communities and Organisations

- Create an organisation for an area or city (for example 'Clean Kukatpally Group').
- Roles: Admin, Moderator, Volunteer. Group page with members, past events and total impact.
- Invite through a link or QR code; announcements and group chat.

D. Cleanup Events

- Create Sunday drives, holiday drives or camping-style cleanups.
- Event details: date, time, meeting point on the map, items to bring (gloves, bags), volunteer limit.
- RSVP with email, SMS and push reminders.
- Event photo gallery with before/after pictures.
- QR check-in at the event and downloadable e-certificates for volunteers.

E. Awareness Section

- Articles and tips: waste segregation (wet, dry, hazardous), composting, plastic-free living.
- Short videos and infographics.
- Monthly challenges such as 'Plastic-Free Week' and 'Clean 5 Spots'.

4. Extra Recommended Features

Feature	Why it helps
Upvote and 'Still dirty' confirmation	Confirms reports and prioritises serious ones
Municipal authority dashboard	Officials see reports by area and mark them resolved
Auto-forward to municipality	Sends complaints by email or city helplines where available
Points, badges, leaderboard	Reporter, Cleaner and Eco Hero levels; top cities and communities
Before/after cleanup proof	Builds trust and motivation
Notifications	Report resolved, new event nearby, group updates
Multi-language support	English, Hindi, Telugu and more for wide adoption
AI garbage detection	Checks the photo really shows garbage and suggests the type
Public city reports	Monthly cleanliness score and area ranking
Spam and fake report control	Report button, moderation queue, user trust score
PWA / mobile friendly	Installable and fast on phones, where most reports come from
Dark/light mode and accessibility	Comfortable for every user
Sponsors and NGOs page	Companies and NGOs can support events

5. Design and Eco-Friendly Look

Colour palette

Forest green	Leaf green	Mint white	Sky blue	Sand cream	Alert red
#2E7D32	#66BB6A	#E8F5E9	#4FC3F7	#F5EBD7	#E53935

Look and feel

- **Backgrounds:** soft sky-to-hill gradient with trees, leaves and clouds; subtle parallax scrolling.
- **Animations:** floating leaves and drifting clouds in the hero; animated counters ('12,450 kg waste cleaned'); a tree that grows as community impact grows; map pins that drop in and turn from red to green with a sparkle when cleaned; smooth scroll reveals.
- **Graphics:** flat illustrations of recycling, Earth, volunteers and bins (SVG or Lottie).
- **Typography:** rounded, friendly fonts such as Poppins or Nunito.
- **Green web practices:** compressed images, lazy loading and a lightweight build so the site itself uses less energy.

6. Suggested Tech Stack

Layer	Options
Frontend	React or Next.js, Tailwind CSS, Framer Motion / Lottie for animation
Maps	Leaflet with OpenStreetMap (free), or Google Maps / Mapbox
Backend	Node.js (Express) or Python (FastAPI / Django)
Database	PostgreSQL with PostGIS for location queries
Authentication	Firebase Auth, or Twilio / MSG91 for phone OTP, plus email login
Media storage	Cloudinary, AWS S3 or Firebase Storage
Notifications	Firebase Cloud Messaging plus email and SMS
Hosting	Vercel / Netlify (frontend), Render / AWS (backend)
AI (later)	Image classification model for garbage detection

7. Database Design (Main Tables)

Table	Key fields
users	id, name, phone, email, city_id, role, points
countries / states / cities	Location hierarchy with ids and names
reports	id, user_id, media_url, lat, lng, address, type, severity, status, upvotes, created_at
report_updates	report_id, before/after photo, status change, updated_by
organizations	id, name, city_id, description, admin_id
org_members	org_id, user_id, role
events	id, org_id, title, date, location, capacity, description
event_rsvps	event_id, user_id, checked_in
comments, badges, notifications	Supporting tables for engagement

8. Development Roadmap

Phase	Deliverables	Time (approx.)
1. MVP	Location selection, OTP/email login, report with photo and map pin, map view	3-4 weeks
2. Community	Organisations, events, RSVP, notifications	3-4 weeks
3. Engagement	Points, badges, leaderboard, awareness section, multi-language	2-3 weeks
4. Authority and scale	Municipality dashboard, auto-forwarding, analytics, AI detection	4+ weeks

9. Risks and Things to Plan For

- **Privacy:** blur faces and vehicle plates, ask for location permission, protect phone numbers.
- **Moderation:** review queue and abuse reporting for inappropriate uploads.
- **Safety:** warn volunteers not to handle hazardous waste (medical, chemical) themselves.
- **Legal:** terms of use and a privacy policy.
- **Cost control:** compress video and set upload limits.
- **Adoption:** partner with local NGOs, colleges and municipal offices to seed the first cities.

> **Next step:** build the Phase 1 MVP - animated landing page, location and login screens, and the interactive garbage map.